

MEDISPHERE CAREPLAN & INTERVENTION

Infosys Springboard Project Report

A Full-Stack Healthcare Management Web Application

Submitted By – Project Team

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Under the Guidance of

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Academic Year: 2026–2027

CERTIFICATE

This is to certify that the project entitled “MediSphere – Careplan & Intervention” is a bona fide project work carried out by _____, bearing Roll Number _____, as part of the Infosys Springboard project program. The project focuses on careplan generation, clinical guideline validation, and patient adherence tracking as an integrated healthcare management workflow.

The work has been completed under the guidance and supervision of the project mentor/guide and is submitted as part of the project documentation requirements.

Role	Name	Signature	Date
Project Guide	_____	_____	_____
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DECLARATION

We hereby declare that the project entitled “MediSphere – Careplan & Intervention” is our original work carried out as part of the Infosys Springboard project. The system was developed under appropriate guidance and supervision. We further declare that the information presented in this report is based on the work carried out during the development of the project and that external resources used for learning and implementation have been acknowledged in the references.

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ABSTRACT

MediSphere – Careplan & Intervention is a full-stack healthcare management module developed as part of the Infosys Springboard project. This milestone extends the MediSphere platform from risk identification and continuous monitoring toward structured clinical planning and intervention support.

The milestone consists of three major modules: AI Careplan Generator, Clinical Guideline Engine, and Adherence Tracking. The AI Careplan Generator uses patient identity and available clinical information such as cardiovascular risk, diabetes risk, blood pressure, HbA1c, medications, and active monitoring alerts to create a personalized careplan containing goals, interventions, tasks, and a planning-oriented predicted outcome.

The Clinical Guideline Engine evaluates patient parameters against demonstration rules. It provides guideline checks for diabetes, HbA1c targets, blood-pressure monitoring, cardiovascular risk, and intervention compatibility, and classifies the result as guideline compliant or review required. The Adherence Tracking module connects the selected patient with a careplan and records completion or missed status for medication, vital monitoring, glucose monitoring, lifestyle, and follow-up tasks. When a patient does not yet have a careplan, the backend can create a default careplan for adherence tracking so the workflow remains usable.

The milestone integrates a React/TypeScript frontend with a Node.js/Express backend and MongoDB persistence. REST APIs connect the careplan, guideline, and adherence workflows. Overall, the milestone demonstrates how patient data can be converted into structured care actions, validated against demonstration guidelines, and monitored through adherence records.

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CHAPTER 1 – INTRODUCTION

1.1 Introduction

Healthcare teams require structured methods to convert patient observations and risk information into actionable care activities. A careplan provides a practical bridge between clinical assessment and ongoing intervention by defining goals, interventions, monitoring tasks, and follow-up actions.

MediSphere Milestone 4 – Careplan & Intervention introduces a coordinated workflow in which patient clinical information is used to generate personalized careplans, validate those plans against demonstration guideline rules, and monitor whether assigned care activities are completed.

1.2 Background

Modern healthcare applications increasingly combine patient data, risk assessment, clinical rules, and digital follow-up workflows. However, risk information alone does not explain what action should be followed or whether the patient is consistently following the recommended activities. A careplan-oriented workflow helps organize these actions and creates a measurable adherence history.

1.3 Problem Statement

Patients may have clinical risk information available in multiple modules without a consistent mechanism for converting that information into a structured careplan. Manual careplan creation can also make it difficult to apply consistent guideline checks and monitor completion of care activities. The milestone therefore requires an integrated approach for personalized careplan generation, guideline validation, and adherence tracking.

1.4 Purpose

The purpose of this milestone is to demonstrate a practical full-stack careplan and intervention workflow that connects patient clinical data with structured goals, interventions, guideline checks, and adherence records through a responsive healthcare management interface.

1.5 Project Information

Item	Details
Project Name	MediSphere – Careplan & Intervention
Milestone	Milestone 04
Program	Infosys Springboard
Project Type	Full-Stack Web Application
Frontend	React.js, TypeScript, HTML5, CSS3
Backend	Node.js, Express.js
Database	MongoDB
Build Tool	Vite
API Style	REST APIs
Major Modules	AI Careplan Generator, Clinical Guideline Engine, Adherence Tracking
Primary Purpose	Structured care planning, guideline validation, and adherence monitoring

CHAPTER 2 – PROJECT OVERVIEW

2.1 Project Description

MediSphere – Careplan & Intervention is designed as a clinical operations support module within the broader MediSphere healthcare platform. It provides a patient-centered workflow for generating careplans, validating clinical guideline conditions, recording care activities, and viewing adherence summaries.

2.2 Major Functional Areas

- Patient selection and clinical profile retrieval
- AI-oriented personalized careplan generation
- Careplan goals and interventions
- Medication and monitoring task generation
- Clinical guideline validation
- HbA1c and diabetes guideline checks
- Blood-pressure monitoring checks
- CVD risk review checks
- Intervention compatibility checks
- Careplan status and approval workflow
- Daily adherence recording
- Adherence summary and category statistics
- Automatic default careplan creation when required for adherence tracking
- REST API and MongoDB persistence

2.3 User Roles

Role	Typical Responsibilities
Administrator	System administration, supported configuration, reports, and permitted management operations
Doctor / Clinician	Review patient information, generate careplans, review guideline results, approve or update careplan status, and monitor adherence
Patient	Access permitted personal careplan information and adherence-related activities where enabled
Medical Staff	Access permitted patient, careplan, and adherence information according to role

CHAPTER 3 – EXISTING SYSTEM

3.1 Existing System

In a conventional workflow, careplan preparation may depend heavily on manual review of patient information. Guideline validation may be performed separately, while adherence may be recorded through disconnected notes or systems. This can make it difficult to maintain a consistent relationship between the patient, careplan, task, and adherence record.

3.2 Problems

- Manual careplan preparation can be time-consuming
- Patient clinical information may be distributed across multiple modules
- Guideline checks may not be consistently documented
- Careplan goals and interventions can be difficult to standardize
- Adherence activities may be tracked separately from the careplan
- Missed or completed tasks may not be summarized automatically
- Patients without an existing careplan may interrupt the adherence workflow
- Status and approval information may not be visible in one workflow

3.3 Need for Improvement

A centralized careplan and intervention workflow can connect patient data, personalized care actions, guideline validation, and adherence tracking. The proposed milestone addresses this need by maintaining patient-linked careplans in MongoDB, exposing REST APIs, applying consistent demonstration rules, and presenting adherence summaries through the frontend.

CHAPTER 4 – PROPOSED SYSTEM

4.1 Proposed Solution

The proposed system provides a full-stack workflow where the frontend communicates with Node.js/Express services through REST APIs. The backend retrieves the selected patient and relevant clinical information, generates a careplan, stores it in MongoDB, validates guideline conditions, and supports adherence record creation and retrieval.

4.2 Key Features

- Patient-linked careplan generation
- Personalized goals based on available clinical indicators
- Medication, vital monitoring, glucose, lifestyle, and follow-up tasks
- Demonstration clinical guideline validation
- Guideline compliance and review-required status
- Careplan status management
- Adherence completion/missed recording
- Automatic default careplan creation when no careplan exists
- Adherence percentage and category-wise statistics
- MongoDB-backed careplan and adherence persistence
- Responsive React/TypeScript interface
- Backend validation and error handling

4.3 Milestone 04 Scope

The milestone concentrates on the three modules defined for Careplan & Intervention: AI Careplan Generator, Clinical Guideline Engine, and Adherence Tracking. It uses available patient and clinical information from the MediSphere platform and is intended as an educational demonstration rather than a clinically validated decision-support product.

CHAPTER 5 – OBJECTIVES AND SCOPE

5.1 Objectives

1. Develop a structured careplan and intervention workflow for clinical operations.
2. Generate personalized careplan goals and interventions from available patient data.
3. Create actionable careplan tasks for medication, monitoring, lifestyle, and follow-up.
4. Validate patient parameters using a consistent demonstration guideline rule set.
5. Provide clear guideline results such as passed, warning, and information statuses.
6. Track whether careplan tasks are completed or missed.
7. Calculate overall and category-wise adherence statistics.
8. Maintain a patient-to-careplan relationship using persistent identifiers.
9. Support careplan status and approval-related workflows.
10. Provide a responsive and user-friendly frontend for careplan and adherence operations.
11. Integrate frontend, backend, and MongoDB through REST APIs.

5.2 Scope

The project covers the design and implementation of a demonstration careplan and intervention platform. It includes patient selection, careplan generation, guideline validation, careplan task creation, adherence recording, summaries, status management, backend APIs, and MongoDB persistence. The implementation is educational and does not replace professional clinical judgment, medical diagnosis, or validated treatment protocols.

CHAPTER 6 – REQUIREMENTS ANALYSIS

6.1 Functional Requirements

Requirement	Description
Patient Retrieval	Retrieve patients available for careplan generation and intervention workflows.
Careplan Generation	Generate a patient-specific careplan from available clinical information.
Goal Generation	Create careplan goals based on blood pressure, HbA1c, CVD risk, diabetes risk, and active alerts where available.
Task Generation	Create medication, vital monitoring, glucose monitoring, lifestyle, and follow-up tasks.
Guideline Validation	Evaluate patient parameters using the demonstration guideline rules.
Guideline Status	Present compliant or review-required outcome with individual check messages.
Adherence Retrieval	Retrieve the latest careplan and associated adherence records for a patient.
Adherence Update	Create or update completed/missed adherence records for a selected task and date.
Careplan Status	Support Draft, Under Review, Approved, Active, and Completed states.
Default Careplan	Create a basic careplan automatically when adherence tracking has no existing careplan for a valid patient.

6.2 Non-Functional Requirements

Requirement	Description
Usability	Interface should be simple and easy to navigate.
Performance	Common careplan and adherence operations should respond efficiently.
Security	Protected application areas should be accessible only to permitted users.
Maintainability	Frontend services and backend routes should remain modular and reusable.
Scalability	Architecture should allow additional careplan rules and intervention categories.
Reliability	Patient, careplan, task, and adherence identifiers should remain consistent.
Responsiveness	The interface should adapt to supported screen sizes.

6.3 Hardware Requirements

- Processor: Intel i3 or equivalent and above
- RAM: 4 GB or above
- Storage: At least 10 GB available for development tools and project files
- Internet connection for package installation and online resources

6.4 Software Requirements

- Windows/Linux/macOS
- Visual Studio Code or equivalent IDE
- Node.js and npm
- MongoDB / MongoDB Atlas
- Modern web browser
- Git/GitHub where used

CHAPTER 7 – TECHNOLOGIES USED

Technology	Purpose
React.js	Frontend user-interface development
TypeScript	Typed frontend application logic and service interfaces
HTML5	Web-page structure
CSS3	Styling and responsive layout
React Router	Client-side navigation
Vite	Frontend development and build tooling
Node.js	Backend runtime environment
Express.js	REST API server and route handling
MongoDB	Persistent storage for patients, careplans, and adherence records
Mongoose	MongoDB schema and model management
JavaScript	Backend application logic
REST APIs	Communication between frontend and backend services
Git/GitHub	Version control and repository management, if used
VS Code	Development environment

7.1 Why These Technologies Were Selected

React and TypeScript support reusable and maintainable healthcare interfaces. Node.js and Express.js provide a lightweight environment for REST APIs and backend workflow logic. MongoDB is suitable for storing patient-linked careplans containing flexible goals, tasks, and adherence records. Mongoose provides schema validation and convenient database access. Vite supports fast frontend development and build operations.

CHAPTER 8 – SYSTEM ARCHITECTURE

8.1 Full-Stack Architecture

The MediSphere Careplan & Intervention platform follows a layered full-stack architecture. The user interacts with the React/TypeScript frontend, which communicates with the Node.js/Express backend through REST APIs. The backend retrieves patient and clinical information, generates and stores careplans, evaluates guideline rules, and manages adherence records in MongoDB.

USER → REACT + TYPESCRIPT FRONTEND → REST API → NODE.JS / EXPRESS BACKEND → CAREPLAN / GUIDELINE / ADHERENCE SERVICES → MONGODB → RESPONSE → REACT UI

[Insert screenshot / diagram here]

Figure 8.1: MediSphere Careplan & Intervention Full-Stack Architecture

8.2 Frontend Architecture

- Pages and route-level views
- Reusable UI components
- Patient selection components
- Careplan generator service
- Guideline validation service
- Adherence tracking service
- Forms and status controls
- Context/state management where used
- Responsive styles and layout

8.3 Backend Architecture

- Express server
- Patient routes
- Careplan routes
- Adherence routes
- Careplan and adherence models
- Patient and clinical data access
- Validation and error handling
- MongoDB/Mongoose integration

8.4 Careplan & Intervention Architecture

- Patient data retrieval
- Clinical data aggregation
- Careplan goal generation
- Careplan task generation
- Careplan persistence
- Guideline rule evaluation
- Guideline status presentation
- Adherence record retrieval
- Completed/missed status update
- Adherence summary calculation

8.5 Data Consistency Architecture

Patient identity is maintained through a patient identifier shared across the relevant services. Careplans store the patient identifier and patient name. Adherence records store the patient identifier, careplan identifier, and task identifier, allowing the system to associate an adherence event with the correct careplan task.

8.6 Security Architecture

The application demonstrates protected application workflows and role-aware access where configured. The milestone is intended for educational demonstration. Production healthcare deployment would require stronger authentication, authorization, encryption, audit logging, privacy controls, secure secrets management, and formal compliance validation.

CHAPTER 9 – SYSTEM DESIGN

9.1 Application Workflow

1. User opens the MediSphere application.
2. User authenticates and accesses the permitted clinical module.
3. User selects a patient.
4. The frontend requests the patient's available information.
5. The backend retrieves patient and relevant clinical data.
6. The AI Careplan Generator creates goals, interventions, and careplan tasks.
7. The generated careplan is stored in MongoDB.
8. The Clinical Guideline Engine evaluates relevant demonstration rules.
9. The user reviews the generated careplan and guideline results.
10. The careplan status can be updated according to the supported workflow.
11. The Adherence Tracking module retrieves the selected patient's careplan.
12. Careplan tasks are displayed for adherence monitoring.
13. The user records a task as completed or missed for a date.
14. The backend stores or updates the adherence record.
15. The dashboard calculates overall and category-wise adherence statistics.

9.2 Use Case Design

The major actors are Administrator, Doctor/Clinician, Patient, and Medical Staff. Their supported use cases include authentication, patient selection, careplan generation, guideline validation, careplan review, status management, adherence tracking, and viewing permitted clinical information.

[Insert screenshot / diagram here]

Figure 9.2: Use Case Diagram – Careplan & Intervention

9.3 Data Flow Design

The high-level data flow is:

Patient → Clinical Data Retrieval → AI Careplan Generator → Goals + Interventions + Tasks → MongoDB → Clinical Guideline Engine → Guideline Results → Adherence Tracking → Completion/Missed Records → Adherence Summary

[Insert screenshot / diagram here]

Figure 9.3: Careplan and Adherence Data Flow Diagram

CHAPTER 10 – SYSTEM MODULES

10.1 AI Careplan Generator

The AI Careplan Generator is responsible for transforming available patient information into a structured careplan. It evaluates indicators such as blood pressure, HbA1c, CVD risk, diabetes risk, medications, and active monitoring alerts. Based on these inputs, it creates careplan goals and intervention statements and builds actionable tasks.

10.2 Clinical Guideline Engine

The Clinical Guideline Engine applies a consistent demonstration rule set to patient data. It evaluates HbA1c, blood pressure, CVD risk, and compatibility-related conditions and produces individual check items with passed, warning, or information status.

10.3 Adherence Tracking

The Adherence Tracking module connects the patient to the latest careplan and displays its tasks. Users can record completed or missed activities. The backend calculates overall adherence and category-wise statistics. If a valid patient does not yet have a careplan, the module can create a default careplan containing standard tracking tasks.

10.4 Careplan Goals and Interventions

Careplan goals are generated according to available clinical triggers. For example, elevated blood pressure can result in an 'Improve blood pressure control' goal, elevated HbA1c can result in an 'Improve glycemic control' goal, and elevated CVD risk can result in a cardiovascular risk reduction goal. Interventions are stored as structured arrays so the frontend can display them consistently.

10.5 Careplan Task Generation

- Medication – follow prescribed medication plan
- Vital Monitoring – blood pressure monitoring
- Glucose Monitoring – glucose monitoring
- Lifestyle – recommended lifestyle activities
- Follow-up – clinical follow-up

10.6 Guideline Check Categories

- Diabetes
- Hypertension
- CVD Risk
- Compatibility

10.7 Guideline Result Status

- Passed – demonstration rule is satisfied
- Warning – the patient parameter requires continued monitoring or review
- Info – informational target or compatibility message

10.8 Careplan Status Management

The backend supports Draft, Under Review, Approved, Active, and Completed states. This allows the careplan workflow to move from generation through review and operational use without changing the underlying patient identity.

10.9 Adherence Record Management

Each adherence record is linked to a patient, careplan, and task. The system uses the task and calendar date to identify an existing record. If a record already exists for the same patient, careplan, task, and day, its status and notes are updated; otherwise, a new record is created.

10.10 Adherence Summary

The dashboard calculates completed records, missed records, total recorded activities, and overall adherence percentage. It also calculates category statistics for Medication, Vital Monitoring, Glucose Monitoring, Lifestyle, and Follow-up.

10.11 Automatic Default Careplan

To prevent an empty adherence workflow, the backend checks whether a careplan exists for the selected patient. When a valid patient has no careplan, a default careplan with standard tasks is created and returned to the adherence dashboard. This ensures that valid patients can still be monitored through the adherence workflow.

10.12 Patient-Careplan Consistency

The system uses the patient identifier as the primary link between the patient collection and careplan collection. Adherence records additionally store the careplan identifier and task identifier. This structure helps prevent an adherence record from being associated with a different patient's careplan.

CHAPTER 11 – FRONTEND IMPLEMENTATION

11.1 React Application Structure

The frontend is implemented using React and TypeScript functional components. The milestone is organized around careplan generation, guideline validation, and adherence tracking views and services.

11.2 Components

- Patient selection and patient summary components
- Careplan generator interface
- Careplan goals and intervention display
- Task list and careplan status controls
- Guideline validation result cards
- Adherence dashboard
- Adherence task controls
- Summary statistics and category indicators
- Forms and validation messages
- Responsive layout components

11.3 Routing

Client-side routing is used to navigate between supported application pages. Protected layouts and role-based rules can be used to restrict access to permitted clinical operations.

11.4 State and Service Layer

Frontend services manage communication with backend APIs and provide the data required by the careplan, guideline, and adherence views. Patient selection is kept consistent across the workflow so that the selected patient remains the reference for generated careplans and adherence records.

11.5 Forms and Validation

The application validates patient selection and supported careplan and adherence actions. Feedback is displayed when a patient is unavailable, a careplan cannot be created, a task is invalid, or an adherence record cannot be saved.

11.6 Careplan Generator UI

The careplan generator displays the selected patient's profile and generated goals, interventions, predicted planning outcome, and approval/status information. The interface is designed to make the generated care actions understandable and reviewable.

11.7 Clinical Guideline UI

The guideline interface presents individual validation checks with category, title, rule description, status, and message. The overall result can be shown as 'GUIDELINE COMPLIANT' or 'REVIEW REQUIRED'.

11.8 Adherence UI

The adherence dashboard presents the active patient, careplan status, careplan tasks, recorded activities, overall adherence, and category-wise adherence. Users can update a task's status as completed or missed.

CHAPTER 12 – BACKEND IMPLEMENTATION

12.1 Backend Overview

The backend is developed using Node.js and Express.js. It provides REST APIs for patient retrieval, careplan generation, careplan status management, guideline-related workflows where applicable, and adherence record management.

12.2 Express Server

The Express server receives HTTP requests, routes them to the appropriate handlers, validates request data, interacts with MongoDB through Mongoose models, and returns structured JSON responses to the frontend.

12.3 Core Routes

- Patient-related routes
- Careplan routes
- Adherence routes
- Clinical data routes used by careplan generation
- Prediction-related routes used as careplan inputs
- Alert-related data used as an optional careplan trigger

12.4 Careplan Generation Backend

The careplan generation endpoint receives a patient identifier, verifies the patient, retrieves latest available clinical information, builds goals and tasks, calculates a planning baseline where applicable, and creates the Careplan document in MongoDB.

12.5 Adherence Backend

The adherence backend retrieves the latest careplan for the requested patient. If none exists, it checks MongoDB for the patient and automatically creates a default careplan. It then retrieves adherence records and calculates summary and category statistics.

12.6 Adherence POST Workflow

1. Validate patientId, careplanId, taskId, date, and status.
2. Validate that status is completed or missed.
3. Validate the careplan ObjectId.
4. Retrieve the careplan.
5. Verify that the careplan belongs to the patient.
6. Verify that the task exists in the careplan.
7. Validate the record date.
8. Find an existing record for the same task and calendar day.
9. Update the existing record or create a new record.
10. Return the saved record as a JSON response.

12.7 Error Handling

The backend returns suitable HTTP status codes and messages for missing patients, missing careplans, invalid identifiers, invalid dates, invalid adherence statuses, missing tasks, and database or server errors.

CHAPTER 13 – DATA MANAGEMENT AND API DESIGN

13.1 Data Management

MongoDB is used to persist patient-linked careplans and adherence records. The Careplan model stores patient identity, generated goals, tasks, predicted planning outcome, status, guideline status, and source clinical values. Adherence records store patient identity, careplan identity, task identity, task details, date, status, and notes.

13.2 Important Data Areas

- Patients
- Patient clinical data
- Careplans
- Careplan goals
- Careplan tasks
- Guideline check results
- Careplan status and approval state
- Adherence records
- Adherence summary statistics
- Clinical alerts used as optional careplan triggers

13.3 API Communication

The frontend communicates with the backend through HTTP requests and REST APIs. The backend validates requests, performs the required business logic, reads or writes MongoDB data, and returns structured JSON responses.

13.4 Core API Areas

Resource	Operation	Purpose
Careplan Patients	GET	Retrieve patients available for careplan generation
Careplans	GET	Retrieve careplans, optionally filtered by patient
Careplan	GET	Retrieve a careplan by its identifier
Careplan Generate	POST	Generate and store a personalized careplan
Careplan Status	PATCH	Update careplan status
Adherence	GET	Retrieve careplan, tasks, records, and adherence statistics for a patient
Adherence	POST	Create or update an adherence record
Adherence Record	DELETE	Delete a selected adherence record

13.5 Careplan Data Structure

A careplan contains patientId, patientName, generatedAt, sourceData, goals, tasks, predictedOutcome, status, and guidelineStatus. Each task contains a taskId, title, category, frequency, instructions, and active state.

13.6 Adherence Data Structure

An adherence record is associated with patientId, careplanId, taskId, task, category, date, status, and notes. The relationship between these identifiers enables the dashboard to retrieve records for the currently selected careplan.

13.7 Default Careplan Data

When a valid patient has no existing careplan, the adherence service can create a default careplan containing a maintenance goal and standard medication, blood pressure, glucose, lifestyle, and follow-up tasks. This prevents valid patients from receiving an unusable 'no careplan found' state.

13.8 Data Consistency

Patient identifiers must match between the Patient, Careplan, and AdherenceRecord collections. The careplan identifier and task identifier provide additional linkage so that a record cannot be intentionally submitted against a different patient's careplan.

CHAPTER 14 – USER INTERFACE AND SCREENSHOTS

Insert the actual screenshots from the final Milestone 04 application in this chapter. Each screenshot should have a figure number and a short caption.

CHAPTER 15 – TESTING

15.1 Testing Overview

Testing was performed to verify patient retrieval, careplan generation, guideline validation, careplan status operations, adherence retrieval, default careplan creation, adherence record updates, and communication between the frontend, backend, and MongoDB.

15.2 Functional Testing

Functional testing verifies that each milestone feature performs its expected operation, including generating careplans from valid patients, displaying guideline results, and recording adherence activities.

15.3 Integration Testing

Integration testing verifies communication between React services, Express routes, Mongoose models, MongoDB collections, and the selected patient workflow.

15.4 UI Testing

UI testing verifies patient selection, careplan display, guideline status presentation, adherence task controls, summary statistics, navigation, validation messages, and responsive layouts.

15.5 Test Cases

ID	Test Case	Expected Result	Status
TC01	Valid patient retrieval	Patient is returned for careplan workflow	Pass
TC02	Invalid patient retrieval	Clear patient-not-found response is displayed	Pass
TC03	Generate careplan	Careplan is created and returned successfully	Pass
TC04	Generate careplan with elevated BP	BP-related goal/intervention is generated	Pass
TC05	Generate careplan with elevated HbA1c	Glycemic-control goal/intervention is generated	Pass
TC06	Generate careplan with high CVD risk	CVD-related goal/intervention is generated	Pass
TC07	Run guideline validation	Guideline checks and overall status are displayed	Pass
TC08	Retrieve adherence for patient with careplan	Tasks and records are displayed	Pass
TC09	Retrieve adherence for patient without careplan	Default careplan is created for valid patient	Pass
TC10	Mark task completed	Adherence record is created/updated	Pass
TC11	Mark task missed	Adherence record is created/updated	Pass
TC12	Update same-day task record	Existing record is updated instead of duplicated	Pass
TC13	Delete adherence record	Selected record is removed	Pass
TC14	Update careplan status	Allowed status is stored successfully	Pass

15.6 Validation and Error Scenarios

Scenario	Expected Behaviour
Missing patientId	Request is rejected with validation message
Unknown patientId	Patient not found response is returned
Invalid careplanId	Invalid identifier response is returned
Careplan belongs to another patient	Adherence operation is rejected
Unknown taskId	Task not found response is returned
Invalid adherence status	Only completed or missed is accepted
Invalid date	Date validation error is returned
Database/server error	HTTP 500 response with failure message is returned

CHAPTER 16 – RESULTS AND DISCUSSION

16.1 Results

The milestone successfully demonstrates an integrated Careplan & Intervention workflow using React/TypeScript, Node.js/Express, and MongoDB. The three milestone modules work together around a common patient identifier and careplan relationship.

16.2 Functional Results

- Patients can be retrieved for careplan generation.
- Personalized careplans can be generated from available clinical information.
- Careplans contain structured goals, interventions, and actionable tasks.
- Clinical guideline rules produce individual validation results.
- Overall guideline status can indicate compliance or review requirement.
- Careplan status can be updated through the supported backend workflow.
- Adherence tracking retrieves the selected patient's careplan and tasks.
- Completed and missed adherence activities can be recorded.
- Existing same-day task records can be updated without creating a duplicate.
- Overall and category-wise adherence statistics are calculated.
- Valid patients without careplans can receive a default careplan for adherence tracking.

16.3 Discussion

The milestone demonstrates the transition from patient assessment and monitoring toward structured intervention management. Careplan generation organizes clinical information into actionable activities, the guideline engine adds a consistent validation layer, and adherence tracking creates measurable feedback about whether activities are being completed.

CHAPTER 17 – CHALLENGES FACED

- Maintaining consistent patient identifiers across frontend and backend modules.
- Ensuring careplans are correctly associated with the selected patient.
- Handling patients who do not yet have a careplan during adherence tracking.
- Creating default careplan tasks without breaking existing generated careplans.
- Keeping task identifiers consistent between careplans and adherence records.
- Preventing duplicate same-day adherence records.
- Handling different clinical data availability for different patients.
- Connecting guideline results to patient-specific values.
- Maintaining consistent UI state when switching between patients.
- Debugging frontend-backend API communication and MongoDB persistence.

17.1 How Challenges Were Addressed

The workflow was organized around `patientId`, `careplanId`, and `taskId`. Backend validation verifies the patient-careplan relationship before saving adherence records. A helper function creates a default careplan when a valid patient has no existing careplan. Same-day records are searched before insertion so that a task can be updated rather than duplicated. Modular services and incremental testing were used to isolate frontend, backend, and database issues.

CHAPTER 18 – LEARNING OUTCOMES

- Understanding React and TypeScript component-based development.
- Creating reusable frontend service functions.
- Working with Node.js and Express.js backend development.
- Creating and consuming REST APIs.
- Understanding MongoDB and Mongoose data models.
- Designing relationships between patient, careplan, task, and adherence data.
- Implementing rule-based clinical guideline checks.
- Building personalized careplan generation logic.
- Creating structured intervention and monitoring tasks.
- Implementing create/update/delete adherence workflows.
- Calculating adherence summaries and category statistics.
- Handling missing careplans through backend fallback logic.
- Testing frontend-backend-database integration.
- Debugging patient-specific data consistency issues.
- Improving technical documentation and project presentation skills.

CHAPTER 19 – ADVANTAGES AND LIMITATIONS

19.1 Advantages

- Centralized careplan and intervention workflow.
- Patient-specific goals and interventions.
- Structured careplan tasks for daily monitoring and follow-up.
- Consistent demonstration guideline validation.
- Clear compliance/review status.
- Patient-linked adherence tracking.
- Automatic default careplan support for valid patients without existing plans.
- Overall and category-wise adherence statistics.
- MongoDB persistence for careplans and adherence records.
- Modular React/TypeScript and Node.js/Express architecture.
- Extendable task categories and guideline rules.

19.2 Limitations

- Guideline rules are demonstration rules and are not clinically validated protocols.
- Careplan generation should be reviewed by qualified healthcare professionals.
- Planning-oriented outcome values are not medically validated predictions.
- Some patient and clinical data may be demonstration or simulated data.
- The system is not a replacement for professional diagnosis or treatment decisions.
- Production deployment would require stronger security, privacy, audit, and compliance controls.
- Real-world clinical deployment would require extensive validation and interoperability testing.

CHAPTER 20 – FUTURE ENHANCEMENTS

1. Integrate validated clinical guideline sources and governance workflows.
2. Add clinician-authored and reusable careplan templates.
3. Introduce configurable intervention protocols by condition.
4. Integrate real-world EHR and FHIR-based patient information.
5. Add advanced predictive models where clinically validated data is available.
6. Use explainable AI methods to show why a careplan recommendation was produced.
7. Add longitudinal adherence trend charts and patient engagement analytics.
8. Add reminder and notification workflows for missed care activities.
9. Support clinician comments, approvals, and careplan version history.
10. Add audit logs for careplan creation, validation, approval, and adherence changes.
11. Strengthen authentication, authorization, encryption, and privacy controls.
12. Deploy the application to a secure cloud environment.
13. Integrate wearable and remote-monitoring data directly into intervention workflows.
14. Support mobile-friendly interfaces for patients and healthcare professionals.

CHAPTER 21 – CONCLUSION

MediSphere – Careplan & Intervention extends the healthcare management platform by providing a structured workflow for turning patient information into actionable care activities. The milestone is centered on three modules: AI Careplan Generator, Clinical Guideline Engine, and Adherence Tracking.

The AI Careplan Generator creates patient-linked goals, interventions, and monitoring tasks from available clinical information. The Clinical Guideline Engine evaluates demonstration rules and provides transparent validation messages. The Adherence Tracking module records completed and missed activities, calculates adherence statistics, and supports automatic default careplan creation when a valid patient does not already have one.

The milestone demonstrates practical integration of React/TypeScript, Node.js/Express, MongoDB, REST APIs, and modular healthcare workflows. The current implementation is intended as an educational and demonstration platform. With validated clinical protocols, real healthcare data, stronger security, interoperability standards, and professional clinical review, the concept can be extended into a more comprehensive care coordination solution.

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- Express.js Documentation – <https://expressjs.com/>
- Vite Documentation – <https://vite.dev/>
- MongoDB Documentation – <https://www.mongodb.com/docs/>
- Mongoose Documentation – <https://mongoosejs.com/docs/>
- MDN Web Documentation – HTML, CSS, and JavaScript references.
- React Router Documentation – Routing concepts and APIs.
- HL7 FHIR Documentation – Healthcare interoperability concepts.
- Infosys Springboard learning resources used during the project.

TEAM MEMBERS AND RESPONSIBILITIES

S. No.	Team Member Name	Roll Number	Role / Responsibility
1	Akshaya Ravichandran	_____	Database / Careplan & Intervention Integration
2	A G Kamalesh	_____	Database
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GROUP PROJECT DECLARATION

We hereby declare that the project entitled “MediSphere – Careplan & Intervention” was carried out collectively by the project team as part of the Infosys Springboard project program. Each team member contributed to the planning, development, testing, integration, and/or documentation of the project according to the assigned responsibilities.

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We also thank our college, classmates, friends, and everyone who supported the team during planning, development, testing, integration, debugging, and documentation.

MILESTONE 04 – FINAL MODULE SUMMARY

Module	Primary Function	Key Output
AI Careplan Generator	Uses patient and clinical information to generate structured care actions	Goals, interventions, tasks, planning outcome
Clinical Guideline Engine	Checks patient parameters against demonstration rules	Passed / Warning / Info checks and overall status
Adherence Tracking	Tracks completion or missed status of careplan activities	Adherence records, overall %, category statistics

END-TO-END MILESTONE FLOW

PATIENT → CLINICAL INFORMATION → AI CAREPLAN GENERATOR → GOALS + INTERVENTIONS + TASKS → CLINICAL GUIDELINE ENGINE → VALIDATION RESULT → CAREPLAN REVIEW / STATUS → ADHERENCE TRACKING → COMPLETED / MISSED RECORDS → ADHERENCE SUMMARY

Milestone 04: Careplan & Intervention

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